Code No: 132AB **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, November/December - 2020
MATHEMATICS-II
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Using Laplace transforms solve y'' – 2y' + y = e^t with y = 2, $\frac{dy}{dt}$ = -1 at t = 0. [15]

2.a) Find $L\left[\frac{1-\cos t}{t}\right]$

b) Find $L^{-1}\left[\frac{s}{(s+1)^2(s^2+1)}\right]$ [7+8]

3.a) Evaluate $\int_0^\infty x^6 e^{-2x} dx$.

b) Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$. [7+8]

4.a) Evaluate $\int_0^{\frac{\pi}{2}} \sin^5\theta \cos^{\frac{7}{2}} d\theta$.

b) Evaluate $\int_0^1 x^7 (1-x^5)^{14} dx$. [7+8]

5.a) Calculate $\iint r^3 \, dr d\theta$ over the area included between the circles r = 2sinθ and r = 4sinθ.

b) Change the order of integration and then evaluate $\int_{-2}^{1} \int_{x^2+4x}^{3x+2} dy dx$. [7+8]

6. Find the volume bounded by the paraboloid $x^2 + y^2 = az$, the cylinder $x^2 + y^2 = 2ay$ and the plane z = 0 (a>0). [15]

7.a) Evaluate i) $\nabla.(3x^2 i + 5xy^2 j + xyz^3 k)$ and ii) $\nabla \times [(xyz)i + (3x^2 y)j + (xz^2 - y^2 z)k]$ at the point (1,2,3).

b) Find the directional derivative of the function $\phi = x^2 yz + 4xz^2$ at $(1,-2,-1)$ in the direction of $2\bar{i} - \bar{j} - \bar{k}$. [8+7]

8. Verify Divergence theorem for $\bar{F}$ =(x^2 – yz) i+(y^2 – zx) j+(z^2 – xy)k, taken over the rectangular parallelepiped 0 ≤ x ≤ a, 0 ≤ y ≤ b, 0 ≤ z ≤ c. [15]

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TUH USED PAPERS OCT/NOV 2